Teach & Repeat Rover Navigation

Project update | Recorded-data simulation

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How we built the recorded-data test

Recorded Freiburg data gave us the route and laser map.

Python parses scans and plans paths. PyBullet drives a Husky rover at 120 Hz.

Recorded route

15 offset starts

Four planners

Bullet runs

All four methods share the same 15 starts, map, controller, and safety checks.

Recorded inputs; simulated repeat runs. ICP remains a separate test.

What was recorded and simulated

Recorded outdoors

Freiburg Campus, July 2004: Pioneer 2AT with a SICK laser.

The raw log contains 15,683 laser scans and odometry poses.

Scans 300–549 give a 29.59 m route; laser returns build the map.

Our repeat trials

Four planners × 15 common starts = 60 Bullet runs.

Offsets −1.2 to +1.2 m; headings −30°, 0°, +30°.

Completion, rejection, contacts, clearance, goal error, and time.

Sensor observations are real. Displaced starts and repeat drives are simulated. Logged odometry is a route reference, not independent ground truth.

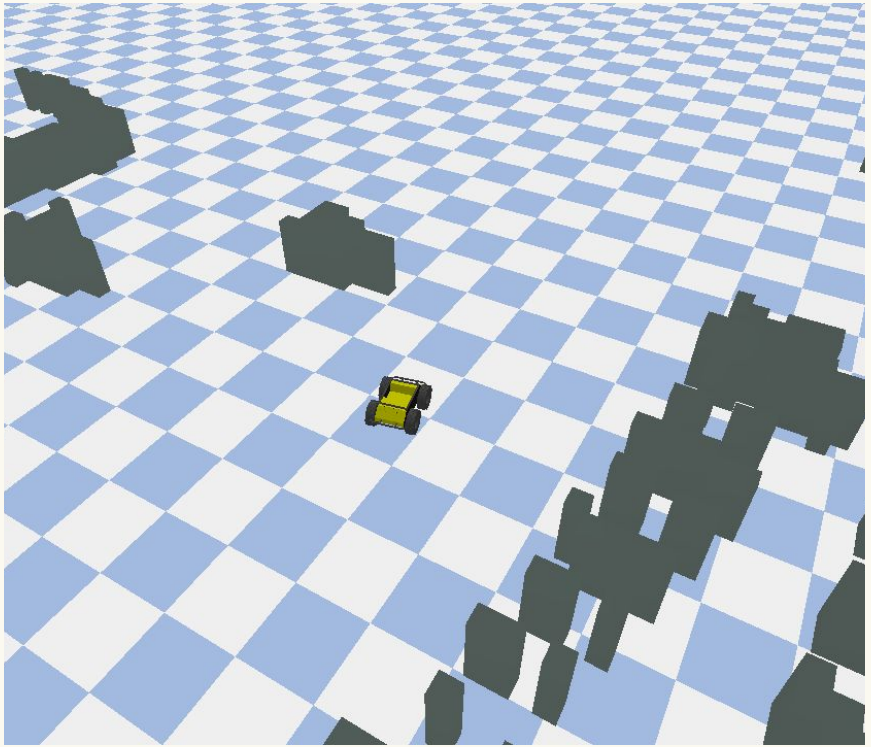
PyBullet rover in the recorded map

The task

Recover a recorded outdoor route from a displaced start while avoiding obstacles seen by the laser.

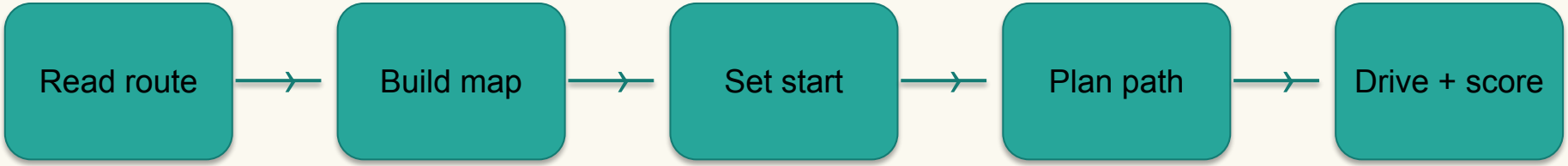
What works today

- 29.59 m logged route
- 751 laser obstacle cells
- Four planners in PyBullet
- Real-scan ICP check



PyBullet Husky surrogate among obstacles mapped from the recorded Pioneer 2AT scans.

From recording to repeat run

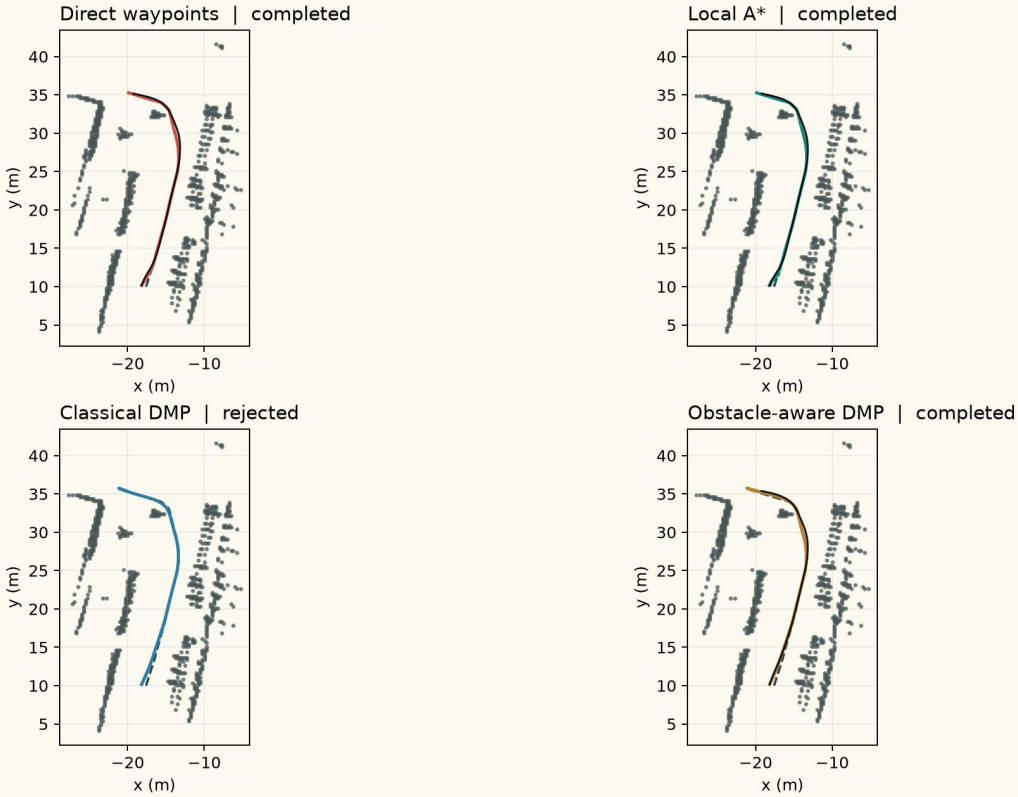


Four candidate paths

1. Direct waypoints · Local A* reconnection
2. Classical DMP · Obstacle-aware DMP

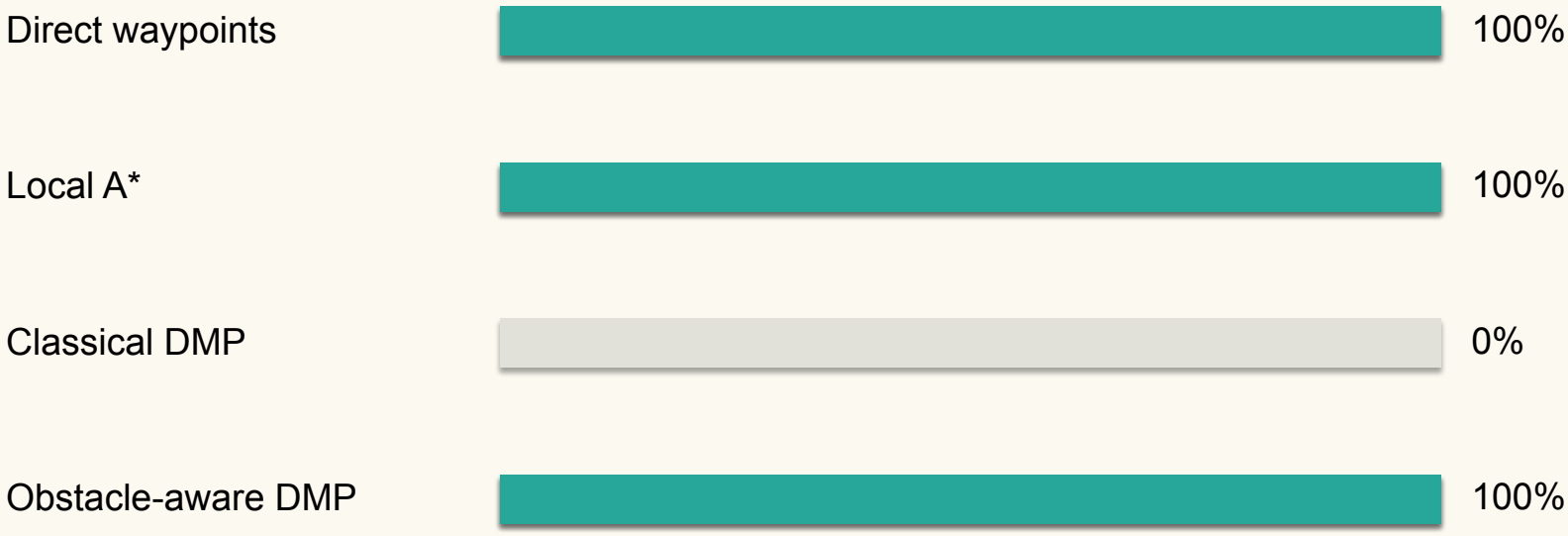
Navigation uses exact simulated pose. The recorded-scan ICP test is separate.

One offset start, four outcomes



Start: +0.6 m lateral, 0° heading. Gray points are recorded laser returns; black traces are Bullet runs.

Completion across 15 starts per method



Three methods completed all 15 starts; classical DMP failed path clearance.

0 obstacle contacts in 60 trials. Rejected paths were not driven.

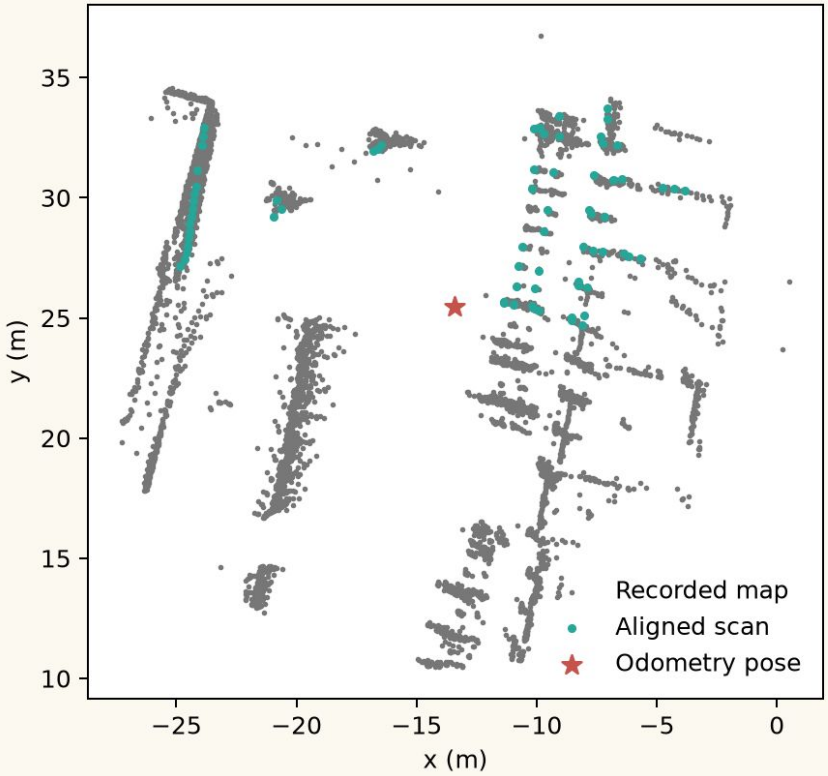
Other measurements from the same runs

Method	Rejected	Stopped	Min. clear.*	Goal err.	Plan time
Direct waypoints	0%	0%	0.287 m	0.646 m	0.12 ms
Local A*	0%	0%	0.287 m	0.646 m	1.59 ms
Classical DMP	100%	0%	N/A	N/A	7.73 ms
Obstacle-aware DMP	0%	0%	0.287 m	0.647 m	10.15 ms

No Bullet obstacle contacts. Goal error is the median among completed runs.

*Grid distance minus a 0.42 m body proxy; timing is planning only, on this PC.

ICP on a recorded laser scan



Held-out Freiburg scan #430

Initial pose offset 0.640 m

After ICP 0.044 m

Map residual 0.047 m

Compared with logged odometry, not ground truth. Navigation still uses exact Bullet pose.

What the results tell us

1. Direct, A*, and coupled DMP completed every start in this recorded segment.

2. Classical DMP failed the 0.15 m path margin; all 15 paths were rejected.

3. No Bullet obstacle contacts occurred with the safety screen enabled.

Next: test other routes and starts; use ICP during navigation; tune the rover model.

Trial design

15 starts = 5 lateral offsets × 3 headings. Route: 29.59 m. Laser map: 0.25 m. Path screen: 0.42 m body radius + 0.15 m margin. PyBullet Husky surrogate at 120 Hz.

Reproduce

python src/recorded_demo.py

Background references

Data: Stachniss & Grisetti, Freiburg Campus 2004 raw log (CC0 via MIT). Software: PyBullet. Code, CSV, plots, and slides: GitHub teach-repeat-rover-navigation.